

Homework #6

Math 742

2026-03-26

Homework: Hypothesis Testing and Likelihood Methods

1. Test statistic, rejection region, and p-value

Let $X_1, \dots, X_n \sim N(\mu, 1)$, with $n = 25$. We wish to test

$$H_0 : \mu = 0 \quad \text{vs.} \quad H_1 : \mu > 0.$$

- a) Construct a test statistic whose distribution is known under H_0 .
- b) Define a rejection region for a level $\alpha = 0.05$ test.
- c) Suppose $\bar{X} = 0.32$. Compute the p-value.
- d) Explain in words what the p-value represents in this context.
- d) If $\alpha = 0.01$, would your conclusion change? Explain.

2. Designing a test via rejection regions

A diagnostic test produces a continuous score $T(X)$. Larger values indicate stronger evidence for disease.

We test

$$H_0 : \text{no disease} \quad \text{vs.} \quad H_1 : \text{disease}.$$

Suppose

$$T(X) \mid H_0 \sim N(0, 1), \quad T(X) \mid H_1 \sim N(2, 1).$$

- a) Describe how to construct a level α test using a rejection region of the form $T > c$.
- b) Express the power at $\mu = 2$ as a function of c .
- c) Explain the tradeoff between Type I and Type II error in this setting.
- d) If the cost of missing a disease is very high, how should c be chosen? Explain.

3. Likelihood ratio test

Let $X_1, X_2, \dots, X_n \sim \text{Exponential}(\lambda)$, where the density is

$$f(x \mid \lambda) = \lambda e^{-\lambda x}, \quad x > 0.$$

We wish to test

$$H_0 : \lambda = 1 \quad \text{vs.} \quad H_1 : \lambda = 2.$$

- a) Write down the likelihood ratio

$$\Lambda(X) = \frac{L(\lambda = 1)}{L(\lambda = 2)}.$$

- b) Show that the likelihood ratio is a monotone function of $\sum_{i=1}^n X_i$.
- c) Based on this, describe the form of the rejection region.
- d) Intuitively explain why small values of $\sum_{i=1}^n X_i$ provide evidence for H_1 .

4. Neyman-Pearson Lemma

Let $X \sim f_\theta(x)$, and consider testing

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta = \theta_1.$$

- a) State the Neyman-Pearson Lemma in words. Do not just copy a formula.
- b) Explain what is meant by a *most powerful test*.
- c) Why does the likelihood ratio appear naturally in the optimal test?
- d) In practice, why is the Neyman-Pearson Lemma limited in scope?
- e) Give an example of a real-world situation where the Neyman-Pearson framework is appropriate.

5. Composite hypotheses and power

Let $X_1, X_2, \dots, X_n \sim N(\mu, 1)$. We wish to test

$$H_0 : \mu = 0 \quad \text{vs.} \quad H_1 : \mu > 0.$$

Consider the test that rejects when

$$\bar{X} > c.$$

- a) Determine c so that the test has level α .
- b) Derive the power function
- $$\beta(\mu) = P_\mu(\text{reject } H_0).$$
- c) Sketch or describe how the power function behaves as μ increases.
- d) Define what it means for this test to be *consistent*.
- e) Show, or argue carefully, whether this test is consistent.